UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO

FACULTAD DE INGENIERÍA

Serie 1

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SEÑALES Y SISTEMAS

Grupo: 1

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Amplitud fase

1. Siendo $a \in \mathbb{C}$, determinar la amplitud y la fase de la señal sinusoidal $x(t) = Acos(\omega_0 t + \phi)$, considerando que $A = |a|$ y $\phi = \measuredangle a, -\pi < \phi \leq \pi$, para los siguientes valores de a:

a) $a = \sqrt{27} + j3$

b) $a = 9\sqrt{2} - j\sqrt{54}$

c) $a = -\sqrt{45} + j\sqrt{15}$

d) $a = -3 - j\sqrt{3}$

$x(t) = A\cos(\omega_0 t + \phi)$; $A = |a|$, $\phi = \measuredangle a$, $-\pi < \phi \leq \pi$

angulo cuya tangente es

Sea $z = x + jy$

$|z| = \sqrt{x^2 + y^2}$

$\measuredangle a = \arg(a) = \theta = \tan^{-1}\left(\frac{y}{a}\right)$

a)

$a = \sqrt{27} + j3$

$|a| = \sqrt{\sqrt{27}^2 + 3^2}$

$= \sqrt{27 + 9}$

$= \sqrt{36}$

$|a| = 6$; $A = 6$

$\phi = \measuredangle a =$

$\arg(a) = \theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\tan^{-1}\left(\frac{3}{\sqrt{27}}\right)$

$a > 0$, $b > 0 \rightarrow$ 1º cuadrante :)

27|3, 9|3, 3|3, 1 $\}3^2$

$\tan^{-1}\left(\frac{\cancel{3}}{3\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{1}{6}\pi$

$\therefore x(t) = 6\cos\left(\omega_0 t + \frac{1}{6}\pi\right)$

b) $a = \overset{a}{9\sqrt{2}} - j\overset{b}{\sqrt{54}}$ $\begin{cases} x = 9\sqrt{2} \\ y = \sqrt{54} \end{cases}$

(-) $+ 2\pi$

$A = |a| = \sqrt{(9\sqrt{2})^2 + \sqrt{54}^2}$

$= \sqrt{81(2) + 54}$

$= \sqrt{162 + 54} = \sqrt{216}$

$A = 14.7$

$\phi = \measuredangle a = \arg(a) = \theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\tan^{-1}\left(\frac{-\sqrt{54}}{9\sqrt{2}}\right) = \frac{-1}{6}\pi$

$-\pi < \phi < \pi$

$+ 2\pi$

$-\frac{1}{6} + \frac{12}{6} = \frac{11}{6}\pi$ queda fuera de rango

$\therefore x(t) = \sqrt{216}\cos\left(\omega_0 t - \frac{1}{6}\pi\right)$

2o cuad. $+\pi$

c) $a = -\sqrt{45} + j\sqrt{15}$ $\begin{cases} x = -\sqrt{45} \\ y = \sqrt{15} \end{cases}$

$\therefore x(t) = \sqrt{60}\cos\left(\omega_0 t + \frac{5}{6}\pi\right)$

$|a| = \sqrt{(-\sqrt{45})^2 + (\sqrt{15})^2}$

$\phi = \theta = \tan^{-1}\left(\frac{y}{x}\right)$

$(-1)^2(\sqrt{45})^2 \rightarrow (1)45$

$= \sqrt{45 + 15}$

$= \sqrt{60}$

$= \tan^{-1}\left(\frac{-\sqrt{15}}{\sqrt{45}}\right) = -\frac{1}{6}\pi + \pi = \frac{5}{6}\pi$

$-\pi < \phi < \pi$

3º Cuadrante

con positivo $+\pi$

d) $a = -3 - j\sqrt{3}$

$$|a| = \sqrt{(-3)^2 + (\sqrt{3})^2}$$
$$= \sqrt{9+3} = \sqrt{12}$$
$$A = \sqrt{12}$$

$$\phi = \tan^{-1}\left(\frac{\sqrt{3}}{-3}\right)$$

$$\frac{1}{6}\pi + \pi = \frac{7}{6}\pi \rightarrow > \pi \therefore \frac{7}{6}\pi - 2\pi = -\frac{5}{6}\pi$$

$$-\pi < \phi < \pi$$

$$\therefore x(t) = \sqrt{12}\cos\left(\omega_0 t - \frac{5}{6}\pi\right)$$

2. Siendo N un entero positivo, demostrar que la secuencia $x[n] = \cos\left(\frac{2\pi}{N}n\right) + j\,\text{sen}\left(\frac{2\pi}{N}n\right)$, es periódica con periodo N. (Es decir, demostrar que $x[n+N] = x[n]\ \forall\, n \in \mathbb{Z}$)

$$x[n] = \cos\left(\frac{2\pi}{N}n\right) + j\,\text{sen}\left(\frac{2\pi}{N}n\right)$$

$$e^{j\Omega_0 n} = \cos(\Omega_0 n) + j\,\text{sen}(\Omega_0 n)$$
$$\Omega_0 = \frac{2\pi}{N}m$$

$$x[n] = e^{j\frac{2\pi}{N}n}$$

$$x[n+N] = x[n]$$
$$x[n+N] = e^{j\frac{2\pi}{N}(n+N)}$$
$$= e^{j\frac{2\pi}{N}n + j\frac{2\pi}{\cancel{N}}\cancel{N}}$$
$$= e^{j\frac{2\pi}{N}n}e^{j2\pi}$$
$$= e^{j\frac{2\pi}{N}n}\left(\overset{1}{\cancel{\cos(2\pi)}} + j\,\overset{0}{\cancel{\text{sen}(2\pi)}}\right)$$

$$x[n+N] = e^{j\frac{2\pi}{N}n} = x[n] \therefore \text{QED}$$

3. Dado un sistema LTI continuo con respuesta al impulso $h(t) = 2\delta(t+3) + 4\delta(t-2)$, mediante convolución determinar la respuesta del sistema a la entrada $x(t) = 5e^{-7t}sen(9t)u(t)$

$$h(t) = 2\delta(t+3) + 4\delta(t-2)$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

Desplazamiento con el impulso unitario desplazado

$$x(t) * \delta(t-t_0) = x(t-t_0)$$

$$x(t) * h(t) = 2[x(t+3)] + 4[x(t-2)]$$

$$x(t+3) = 5e^{-7(t+3)}\text{sen}[9(t+3)]u(t+3)$$
$$x(t-2) = 5e^{-7(t-2)}\text{sen}(9(t-2))u(t-2)$$

$$y(t) = 10e^{-7(t+3)}\text{sen}[9t+27])u(t+3) + 20e^{-7(t-2)}\text{sen}(9t-18))u(t-2)$$

4. Determinar $f[n] = x[n] * y[n]$ para las siguientes secuencias

a) $x[n] = -\delta[n] + 2\delta[n-1] - 3\delta[n-2] - 4\delta[n-3]$

$y[n] = \delta[n] - 2\delta[n-2]$

b) $x[n] = \{1, -1, 4, 2\}$, $y[n] = \{1, -1, 2\}$

(B)
$x[n] * \delta[n-n_0]$
$= x[n-n_0]$

(A)
$(\alpha x_1) * (\beta x_2) = \alpha\beta(x_1 * x_2)$

a) $f[n] = x[n] * y[n]$

Sust. y

$= x[n] * (\delta[n] - 2\delta[n-2])$

Por distributividad

$= x[n] * \delta[n] - x[n] * 2\delta[n-2]$

por asociatividad con el escalamiento en amplitud ...A

$= x[n] * \delta[n] - 2(x[n] * \delta[n-2])$

por desplazamiento con el impulso unitario desplazado ...B

$= x[n] - 2(x[n-2])$

Sustituyendo x

$= -\delta[n] + 2\delta[n-1] - 3\delta[n-2] - 4\delta[n-3]$
$+ 2\delta[n-2] - 4\delta[n-1-2] + 6\delta[n-2-2] + 8\delta[n-3-2]$

$= -\delta[n] + 2\delta[n-1] - 3\delta[n-2] - 4\delta[n-3]$
$+ 2\delta[n-2] - 4\delta[n-3] + 6\delta[n-4] + 8\delta[n-5]$

$= -\delta[n] + 2\delta[n-1] - \delta[n-2] - 8\delta[n-3] + 6[n-4] + 8\delta[n-5]$

b) $x[n] = \{1, -1, 4, 2\}$ $\quad y[n] = \{1, -1, 2\}$

$L_x = 4$ $\quad L_y = 3$ $\quad L_f = L_x + L_y - 1$

$L_f = 6 \quad 0 \le n \le 5$

$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]\, y[n-k]$$

$$f[n] = \sum_{k=0}^{L_n - 1} x[k]\, y[n-k]$$

$f[0] = (1, -1, 4, 2) \cdot (1, 0, 0, 0) = 1$
$f[1] = (1, -1, 4, 2) \cdot (-1, 1, 0, 0) = -1 - 1 = -2$
$f[2] = (1, -1, 4, 2) \cdot (2, -1, 1, 0) = 7$
$f[3] = (1, -1, 4, 2) \cdot (0, 2, -1, 1) = -4$
$f[4] = (1, -1, 4, 2) \cdot (0, 0, 2, -1) = 6$
$f[5] = (1, -1, 4, 2) \cdot (0, 0, 0, 2) = 4$

$f[n] = \{1, -2, 7, -4, 6, 4\}$

5. Dado el sistema discreto LTI IIR, representado mediante la ecuación en diferencias

$$y[n] - 4y[n-1] - 3y[n-2] = 11x[n-1] - 5x[n-2]$$

Considerando $x[n] = \{4, 1, -3, -2\}$, obtener $y[n]$ para $n = 0, 1, 2, 3$.

Reescribimos $y[n]$ en su forma recursiva

$$y[n] = 11x[n-1] - 5x[n-2] + 4y[n-1] + 3y[n-2]$$

como es LTI, CI = 0; $y[a] = 0 \ \forall\, a < 0$

Teniendo $x[n] = \{\underset{0}{4}, \underset{1}{1}, \underset{2}{-3}, \underset{3}{-2}\}$

$n = 0$

$$y[0] = 11\cancel{x[-1]} - 5\cancel{x[-2]} + \cancel{4y[-1]} + \cancel{3y[-2]}$$
$$= 0$$

$n = 1$

$$y[1] = 11x[0] - 5\cancel{x[-1]} + 4y[0] + \cancel{3y[-1]}$$
$$= 11(4) + \cancel{4(0)}$$
$$= 44$$

$n = 2$

$$y[2] = 11x[1] - 5x[0] + 4y[1] + 3y[0]$$
$$= 11 - 5(4) + 4(44) + \cancel{3(0)}$$
$$= -20 + 176 + 11$$
$$= 167$$

$n = 3$

$$y[3] = 11x[2] - 5x[1] + 4y[2] + 3y[1]$$
$$= 11(-3) - 5 + 4(167) + 3(44)$$
$$= -33 - 5 + 668 + 132$$
$$-38$$
$$= 762$$

$$\begin{array}{r} 167 \\ \times 4 \\ \hline 668 \\ 132 \\ 800 \end{array}$$

$$y[n] = \{0, 44, 167, 762\}$$

6. Dado el sistema discreto LTI IIR, representado mediante la ecuación en diferencias

$$y[n] - 2y[n-1] + y[n-2] - 3y[n-3] = x[n] - 3x[n-2]$$

Obtener la respuesta del sistema a $x[n] = \{\overset{0}{1}, \overset{1}{3}, \overset{2}{1}, \overset{3}{7}, \overset{4}{4}\}$, para $n = 0, 1, 2, 3, 4$

Se reescribe la ec. en diferencias en su forma recursiva

$$y[n] = x[n] - 3x[n-2] + 2y[n-1] - y[n-2] + 3y[n-3]$$

Como es LTI, se consideran $CI = 0$: $y[-1] = y[-2] = \ldots = y[-N] = 0$

$n=0$

$$y[0] = x[0] - \cancel{3x[-2]} + \cancel{2y[-1]} - \cancel{y[-2]} + \cancel{3y[-3]}$$

$$= 1$$

$n=1$

$$y[1] = x[1] - \cancel{3x[1-2]} + 2y[1-1] - \cancel{y[1-2]} + \cancel{3y[1-3]}$$

$$= 3 + 2(1)$$

$$= 5$$

$n=2$

$$y[2] = x[2] - 3x[0] + 2y[2-1] - y[0] + \cancel{3y[2-3]}$$

$$= 1 - 3 + 2(5) - 1$$

$$= 7$$

$n=3$

$$y[3] = x[3] - 3x[1] + 2y[2] - y[1] + 3y[0]$$

$$7 - 3(3) + 2(7) - 5 + 3$$

$$= 10$$

$n=4$

$$y[4] = x[4] - 3x[2] + 2y[3] - y[2] + 3y[1]$$

$$= 4 - 3(1) + 2(10) - 7 + 3(5)$$

$$4 - 3 + 20 - 7 + 15$$

$$= 29$$

$$y[n] = \{1, 5, 7, 10, 29\}$$

7. Dado un sistema LTI continuo con respuesta al impulso $h(t) = e^{-2t}\cos(\sqrt{3}\,t)u(t)$, obtener la ecuación diferencial que también modela al sistema en el dominio del tiempo t.

$x \to \boxed{\begin{matrix}EDL\\ CI=0\end{matrix}} \to y \qquad x \to \boxed{H(s)} \to Y$

$$\mathcal{L}\{h(t)\} = \mathcal{L}\{H(s)\}$$

$$\mathcal{L}\{e^{-2t}\cos(\underset{\omega_0}{\sqrt{3}t})u(t)\} = \frac{s+2}{(s+2)^2+(\sqrt{3})^2} = \frac{s+2}{(s+2)^2+3} = H(s)$$

$$\frac{Y(s)}{X(s)} = \frac{s+2}{(s+2)^2+3}$$

$$((s+2)^2+3)\,Y(s) = X(s)\,(s+2)$$

$$\downarrow$$

$$s^2+4s+4+3$$

$$\downarrow$$

$$(s^2+4s+7)\,Y(s) = \ldots$$

$$\mathcal{L}^{-1}\{s^2Y(s)+4sY(s)+7Y(s) = sX(s)+2X(s)\}$$

$$\frac{d^2y(t)}{dt^2} + \frac{4\,dy(t)}{dt} + 7y(t) = \frac{dx(t)}{dt} + 2x(t)$$

8. Dado un sistema LTI discreto con resp. al impulso $h[n] = 0.3^n u[n] - 0.7^n u[n] + 0.2^n u[n]$, obtener la ecuación en diferencias que también modela al sistema en el dominio del tiempo n.

$$\mathcal{Z}\{h[n]\} = H(z)$$

$$H(z) = \frac{Y(z)}{X(z)};$$

$$\mathcal{Z}\{h[n]\} = \left(\frac{z}{z-0.3} - \frac{z}{z-0.7} + \frac{z}{z-0.2}\right)$$

$$\downarrow$$

$$\frac{z(z-0.7)-z(z-0.3)}{(z-0.3)(z-0.7)} + \frac{z}{z-0.2}$$

$$\frac{z(z-0.7)(z-0.2) - z(z-0.3)(z-0.2) + z(z-0.3)(z-0.7)}{(z-0.3)(z-0.7)(z-0.2)}$$

$$P(z): (z^2-0.7z)(z-0.2) - (z^2-0.3z)(z-0.2) + (z^2-0.3z)(z-0.7)$$

$$= \cancel{z^3} - \cancel{0.2z^2} - 0.7z^2 + 0.14z - \cancel{z^3} + \cancel{0.2z^2} + \cancel{0.3z^2} - 0.06z + z^3 - 0.7z^2 - \cancel{0.3z^2} + 0.21z$$

$$= z^3 - 1.4z^2 + 0.29z$$

$$Q(z): (z^2 - 0.7z - 0.3z + 0.21)(z - 0.2)$$
$$= (z^2 - z + 0.21)(z - 0.2)$$
$$= z^3 - 0.2z^2 - z^2 + 0.2z + 0.21z - 0.042$$
$$= z^3 - 1.2z^2 + 0.41z - 0.042$$

$$\therefore \frac{Y(z)}{X(z)} : \frac{z^3 - 1.4z^2 + 0.29z}{z^3 - 1.2z^2 + 0.41z - 0.042}$$

$$(z^3 - 1.2z^2 + 0.41z - 0.042)\, Y(z) = X(z)(z^3 - 1.4z^2 + 0.29z)$$

$\rightarrow$ Dividir entre z^3 para hacerlo causal (forma z^{-1})

$$(1 - 1.2z^{-1} + 0.41z^{-2} - 0.042z^{-3})\, Y(z) = X(z)(1 - 1.4z^{-1} + 0.29z^{-2})$$
$$\mathcal{Z}^{-1}\left\{ Y(z) - (1.2z^{-1})Y(z) + (0.41z^{-2})Y(z) - (0.042z^{-3})Y(z) = X(z) - (1.4z^{-1})X(z) + (0.29z^{-2})X(z) \right\}$$

$$\rightarrow y[n] - 1.2\, y[n-1] + 0.4\, y[n-2] - 0.042\, y[n-3] = x[n] - 1.4\, x[n-1] + 0.29\, x[n-2]$$

9. Dado el sistema LTI causal representado por la siguiente EDL de orden N=2.

$$\mathcal{Z}\left\{ \frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = 5\frac{d^2x(t)}{dt^2} + 16\frac{dx(t)}{dt} + 18x(t) \right\}$$

Obtener $H(s)$, determinar si el sistema es estable, y obtener la respuesta del sistema a $x(t) = u(t)$.

$$s^2 Y(s) + 5s\, Y(s) + 6\, Y(s) = 5s^2 X(s) + 16s\, X(s) + 18\, X(s)$$
$$(s^2 + 5s + 6)\, Y(s) = (5s^2 + 16s + 18)\, X(s)$$
$$H(s) = \frac{Y(s)}{X(s)} = \frac{5s^2 + 16s + 18}{s^2 + 5s + 6}$$

Analizamos P(Q) para determinar estabilidad

$$\frac{-5 \pm \sqrt{25 - 24}}{2} = \frac{-5 \pm 1}{2} = \begin{cases} P_1: -2 \\ P_2: -3 \end{cases} \qquad (s+2)(s+3)$$

como $Re\{P_{1,2}\} < 0$ entonces el sistema es estable.

Respuesta al sistema $x(t) = u(t)$

$$Y(s) = H(s)X(s), \quad \mathcal{Z}\{u(t)\} = \frac{1}{s} \quad x(s)$$

$$\therefore Y(s) = \frac{5s^2 + 16s + 18}{s(s+2)(s+3)} \quad \begin{matrix} gr: 2 \\ gr: 3 \end{matrix}$$ $\rightsquigarrow$ se procede a hacer expansión en fracciones parciales

$$\frac{5s^2 + 16s + 18}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3}$$

$$5s^2+16s+18 = A(s+2)(s+3) + B(s)(s+3) + C(s)(s+2)$$

Si $s=-2$

$$20-32+18 = B(-2)$$
$$+6 = B(-2)$$
$$B = \frac{6}{-2} = -3$$

Si $s=0$

$$18 = A(2)(3)$$
$$18 = A\,6$$
$$A = \frac{18}{6}$$
$$A = 3$$

Si $s=-3$

$$45-48+18 = C(-3)(-1)$$
$$-3+18 = 3C$$
$$15 = 3C$$
$$C = \frac{15}{3} = 5$$

$\therefore$ $A=3$, $B=-3$, $C=5$

$$\therefore \quad Y(s) = \frac{3}{s} - \frac{3}{s+2} + \frac{5}{s+3}$$

Aplicando $\mathcal{L}^{-1}$

$$y(t) = 3u(t) - 3e^{-2t}u(t) + 5e^{-3t}u(t)$$

$$y(t) = \left(3 - 3e^{-2t} + 5e^{-3t}\right)u(t)$$

10. Dado el sistema LTI causal representado por la siguiente EDL de orden N=2.

$$\mathcal{L}\left\{\frac{d^2y(t)}{dt^2} - 4\frac{dy(t)}{dt} + 4y(t) = x(t)\right\}$$

Obtener $H(s)$, determinar si el sistema es estable, y obtener su respuesta a $x(t) = 9e^{-t}u(t)$.

$$s^2Y(t) - 4sY(t) + 4Y(t) = X(t)$$
$$(s^2-4s+4)Y(t) = X(t)$$
$$\frac{Y(t)}{X(t)} = \frac{1}{s^2-4s+4} = H(s)$$
$$= \frac{1}{(s-2)^2}$$

$$Q(s) = s^2-4s+4$$

roots

$$P_{1,2} = 2 \qquad \mathrm{Re}(P_{1,2}) > 0$$

$\therefore$ NO es estable

$$Y(s) = H(s)X(s)$$

$$\mathcal{L}\{x(t)\} = \mathcal{L}\{9e^{-t}u(t)\} = 9\frac{1}{s+1}$$

$$\therefore \quad Y(s) = \frac{9}{(s+1)(s-2)^2} \quad \begin{matrix} gr:0 \\ gr:3 \end{matrix}$$

✓ procedemos con expansión en fracciones parciales

BCE

$$\frac{9}{(s+1)(s-2)^2} = \frac{A}{(s+1)} + \frac{B}{(s-2)^2} = \frac{A}{s+1} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$$

$$9 = A(s-2)^2 + B(s+1)(s-2) + C(s+1)$$

Si $s=2$

$9 = C(3)$

$C = \frac{9}{3} = 3$

Si $s=0$

$9 = A(-2)^2 + B(1)(-2) + C$

$9 = 4 - 2B + 3$

$2 = -2B$

$B = \frac{2}{-2} = -1$

Si $s=-1$

$9 = A(-3)^2$

$9 = A(9)$

$A = 1$

$A = 1$

$B = -1$

$C = 3$

$$\therefore Y(s) = \frac{1}{s+1} - \frac{1}{s-2} + \frac{3}{(s-2)^2}$$

Aplicamos $\mathcal{L}^{-1}$

$$y(t) = e^{-t}u(t) - e^{2t}u(t) + 3te^{2t}u(t)$$

$$y(t) = \left(e^{-t} - e^{2t} + 3te^{2t}\right)u(t)$$

11. Dado el sistema LTI causal representado por la siguiente EDL de orden N=2.

$$\mathcal{L}\left\{ \frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 2y(t) = x(t) \right\}$$

Obtener $H(s)$, determinar si el sistema es estable, y obtener la respuesta del sistema a $x(t) = u(t)$.

$$= s^2 Y(s) + 2sY(s) + 2Y(s) = X(s)$$

$$(s^2+2s+2)Y(s) = X(s)$$

$$\frac{Y(s)}{X(s)} = H(s) = \frac{1}{s^2+2s+2}$$

$Q(s) = s^2+s+2$

$$\frac{-2 \pm \sqrt{4-8}}{2}$$

$P_1 = \frac{-\cancel{8} + \sqrt{6}j}{2}$

$P_2 = \frac{-2 - \sqrt{6}j}{2}$

Como $Re\{P_{1,2}\} < 0$

el sistema es estable

$$x(t) \rightarrow \mathcal{L}\{x(t)\} = \frac{1}{s}$$

$$Y(s) = H(s)X(s) = \frac{1}{s(s^2+2s+2)}$$

$\rightarrow$ se procede con expansión en fracciones parciales

$$\frac{1}{s(s^2+2s+2)} = \frac{A}{s} + \frac{Bs+C}{s^2+2s+2}$$

$1 = A(s^2+2s+2) + (Bs+C)s$

Si $s=0$

$1 = A(2)$

$A = \frac{1}{2}$

Si $s=1$

$1 = A(1+2+2) + (B+C)$

$1 = \frac{5}{2} + (B+C)$

$B+C = -\frac{3}{2}$

Si $s=-1$

$1 = A(1\cancel{-2+2}) + (-B+C)(-1)$

$1 = A + (B-C)$

$B - C = \frac{1}{2}$

$$\begin{aligned} B+\cancel{C} &= -\tfrac{3}{2} \\ B\cancel{-C} &= \tfrac{1}{2} \\ \hline 2B &= -1 \\ B &= -\tfrac{1}{2} \end{aligned}$$

$-\frac{1}{2} - C = \frac{1}{2}$

$-C = 1$

$C = -1$

$A = \frac{1}{2}$

$B = -\frac{1}{2}$

$C = -1$

$$Y(s) = \frac{1}{2s} + \frac{(-\frac{1}{2})s - 1}{s^2+2s+2}$$

$$\mathcal{L}^{-1}\left\{\frac{(-\frac{1}{2})s - 1}{s^2+2s+2}\right\} =$$

$$e^{-t}\left[-\frac{1}{2}\cos(t) - \frac{1}{2}\operatorname{sen}(t)\right]u(t)$$

$a = 1$

$\omega_0 = 0.5\sqrt{4(2)-4}$

$= 0.5\sqrt{4}$

$\omega_0 = 1$

$A_1 = -\frac{1}{2}$

$A_2 = \frac{2(-1) - (-\frac{1}{2})(2)}{2}$

$= \frac{-2+1}{2} = -\frac{1}{2}$

Aplicando $\mathcal{L}^{-1}$:

$$y(t) = \frac{1}{2}u(t) - \frac{1}{2}e^{-t}[\cos(t)+\operatorname{sen}(t)]u(t)$$

$$y(t) = \frac{1}{2}\left(1 - e^{-t}[\cos(t)+\operatorname{sen}(t)]\right)u(t)$$

12. Dado el sistema LTI causal representado por la siguiente EDL de orden N=2.

$$\mathcal{Z}\left\{ y[n] - 0.75y[n-1] + 0.125y[n-2] = x[n-1] \right\}$$

Obtener $H(z)$, determinar si el sistema es estable, y obtener su respuesta a $x[n] = (0.5)^n\, u[n]$.

$$Y(z) - 0.75 z^{-1} Y(z) + 0.125 z^{-2} Y(z) = z^{-1} X(z)$$

$$\frac{Y(z)}{X(z)} = \boxed{H(z) = \frac{z^{-1}}{1 - 0.75z^{-1} + 0.125z^{-2}}} \quad \frac{z^2}{z^2}$$

$$= \frac{z}{z^2 - 0.75z + 0.125} \rightsquigarrow z^2 - \frac{3}{4}z + \frac{1}{8}$$

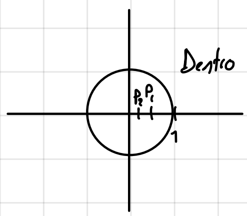
$$Q(z): z^2 - 0.75z + 0.125 \qquad \frac{0.125}{1.000}\ 8$$

$$z^2 - \frac{3}{4} + \frac{1}{8}$$

$$\frac{\frac{3}{4} \pm \sqrt{\frac{9}{16} - \frac{1}{2}\left(\frac{8}{8}\right)}}{2} = \frac{\frac{3}{4} \pm \sqrt{\frac{1}{16}}}{2} = \frac{\frac{3}{4} \pm \frac{1}{4}}{\frac{2}{1}} \begin{cases} \frac{1}{2} \\ \frac{1}{4} \end{cases}$$

$$P_1 = \frac{1}{2} \qquad P_2 = \frac{1}{4}$$

$$(z - \tfrac{1}{2})(z - \tfrac{1}{4})$$



Dentro del círculo unitario

$\therefore$ es estable

pues $|P_1|$ y $|P_2| < 1$

para respuesta a $x[n] = (0.5)^n u[n]$

$$\mathcal{Z}\{x[n]\} = \frac{z}{z - 0.5} = X(z)$$

$$Y(z) = H(z)\, X(z)$$

$$Y(z) = \frac{z}{(z^2 - 0.75z + 0.125)} \left(\frac{z}{z - 0.5}\right) = \frac{z\,(z)}{(z - \frac{1}{4})(z - \frac{1}{2})^2}$$

$$\frac{Y(z)}{z} = \frac{z}{(z - \frac{1}{4})(z - \frac{1}{2})^2}$$

$$\frac{z}{(z - \frac{1}{2})(z - \frac{1}{4})(z - \frac{1}{2})} = \frac{A}{z - \frac{1}{4}} + \frac{B}{z - \frac{1}{2}} + \frac{C}{(z - \frac{1}{2})^2}$$

$$z = A\left(z-\frac{1}{2}\right)^2 + B\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right) + C\left(z-\frac{1}{4}\right)$$

Si $z=\frac{1}{2}$

$$\frac{1}{2} = C\left(\frac{1}{2}-\frac{1}{4}\right)$$

$$\frac{1}{2} = C\left(\frac{1}{4}\right)$$

$$C = \frac{\frac{1}{2}}{\frac{1}{4}} = \frac{4}{2} = 2$$

Si $z=\frac{1}{4}$

$$\frac{1}{4} = A\left(\frac{1}{4}-\frac{1}{2}\right)^2$$

$$\frac{1}{4} = A\left(-\frac{1}{4}\right)^2$$

$$\frac{1}{4} = A\left(\frac{1}{16}\right)$$

$$A = \frac{\frac{1}{4}}{\frac{1}{16}} = \frac{16}{4} = 4$$

Si $z=1$

$$1 = 4\left(1-\frac{1}{2}\right)^2 + B\left(1-\frac{1}{2}\right)\left(1-\frac{1}{4}\right) + 2\left(1-\frac{1}{4}\right)$$

$$1 = 4\left(\frac{1}{4}\right) + B\left(\frac{1}{2}\right)\left(\frac{3}{4}\right) + \frac{6}{4}$$

$$1 = 1 + \frac{3}{8}B + \frac{6}{4}$$

$$-\frac{6}{4} = \frac{3}{8}B$$

$$B = \frac{-\frac{6}{4}}{\frac{3}{8}} = \frac{-48}{12} = -4$$

$A=4$
$B=-4$
$C=2$

$\therefore$

$$\frac{Y(z)}{z} = \frac{4}{z-\frac{1}{4}} - \frac{4}{z-\frac{1}{2}} + \frac{2}{\left(z-\frac{1}{2}\right)^2}$$

$$Y(z) = 4\,\frac{z}{z-\frac{1}{4}} - 4\,\frac{z}{z-\frac{1}{2}} + 2\,\underbrace{\frac{z}{\left(z-\frac{1}{2}\right)^2}}$$

Para la forma $\frac{a\,z}{(z-a)^2}$ $\quad a=\frac{1}{2}$ $\quad \frac{1}{\frac{1}{2}}\;\frac{\frac{1}{2}z}{\left(z-\frac{1}{2}\right)^2}$

Aplicando $\mathcal{Z}^{-1}$

$$y[n] = 4\left(\frac{1}{4}\right)^n u[n] - 4\left(\frac{1}{2}\right)^n u[n] + 4\,n\left(\frac{1}{2}\right)^n u[n]$$

$$y[n] = 4\left(\left(\frac{1}{4}\right)^n - \left(\frac{1}{2}\right)^n + n\left(\frac{1}{2}\right)^n\right)u[n]$$

13. Dado el sistema LTI causal representado por la siguiente EDL de orden N=1

$$\mathcal{Z}\left\{ 2y[n] - y[n-1] = 2x[n] \right\}$$

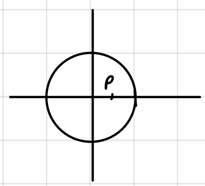
Obtener $H(z)$, determinar si el sistema es estable o no, y obtener la respuesta del sistema a $x[n] = sen(\frac{\pi}{3}n)u[n]$.

$$2Y(z) - z^{-1}Y(z) = 2X(z)$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{2}{2 - z^{-1}} \cdot \left(\frac{z}{z}\right) = \frac{2z}{2z-1} = 2\left(\frac{z}{z - \frac{1}{2}}\right)$$

$$Q(z) = 2z - 1$$

$$P = z = \frac{1}{2}$$



Dentro del Circ Unitario

$\therefore$ es estable

Para la salida de acuerdo a la entrada

$$x[n] = sen\left(\frac{\pi}{3}n\right)u[n] \quad \begin{cases} r = 1 \\ \Omega_0 = \frac{\pi}{3} \end{cases}$$

$$\mathcal{Z}\left\{x[n]\right\} = X(s) = \frac{\left(sen\frac{\pi}{3}\right)z^{-1}}{1 - \left(2\cos\frac{\pi}{3}\right)z^{-1} + z^{-2}} \left(\frac{z^2}{z^2}\right)$$

$$= \frac{z\, sen\left(\frac{\pi}{3}\right)}{z^2 - z\left(2\cos\frac{\pi}{3}\right) + 1}$$

$$Y(z) = 2\left(\frac{z\, sen\left(\frac{\pi}{3}\right)}{z^2 - z\left(2\cos\frac{\pi}{3}\right) + 1}\right)\left(\frac{z}{z - \frac{1}{2}}\right)$$

$$sen\frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\cos\frac{\pi}{3} = \frac{1}{2}$$

$$\frac{Y(z)}{z} = \frac{\cancel{2}\, z\, \frac{\sqrt{3}}{\cancel{2}}}{\left(z^2 - z + 1\right)\left(z - \frac{1}{2}\right)}$$

$$\frac{Y(z)}{z} = \sqrt{3}\, \underbrace{\frac{z}{\left(z - \frac{1}{2}\right)\left(z^2 - z + 1\right)}}$$

→ propia, procedemos a expandirla por fracciones parciales

$$\frac{z}{\left(z - \frac{1}{2}\right)\left(z^2 - z + 1\right)} = \frac{A}{z - \frac{1}{2}} + \frac{Bz + C}{z^2 - z + 1}$$

$$z = A(z^2 - z + 1) + (Bz + C)\left(z - \frac{1}{2}\right)$$

Si $z = \frac{1}{2}$

$$\frac{1}{2} = A\left(\frac{1}{4} - \frac{1}{2} + 1\right)$$

$$\frac{1}{2} = A\left(\frac{3}{4}\right)$$

$$A = \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{4}{6} = \frac{2}{3}$$

Si $z = 1$

$$1 = A(1 - 1 + 1) + (B + C)\left(1 - \frac{1}{2}\right)$$

$$1 = A + \frac{1}{2}(B + C)$$

$$\frac{1}{3} = \frac{1}{2}B + C$$

$$B + C = \frac{2}{3}$$

Si $z = -1$

$$-1 = A(1 + 1 + 1) + (-B + C)\left(-1 - \frac{1}{2}\right)$$

$$-1 = \cancel{3}\frac{2}{\cancel{3}} + (-B + C)\left(-\frac{3}{2}\right)$$

$$-3 = (-B + C)\left(\frac{-3}{2}\right)$$

$$-B + C = \frac{\cancel{-3}(2)}{\cancel{-3}}$$

$$B - C = -2$$

$$B + \cancel{C} = \frac{2}{3}$$

$$B - \cancel{C} = -2$$

$$2B = -\frac{4}{3}$$

$$B = -\frac{\cancel{8}}{3}$$

$$-\frac{2}{3} + C = \frac{2}{3}$$

$$C = \frac{4}{3}$$

$$A = \frac{2}{3}$$

$$B = -\frac{2}{3}$$

$$C = \frac{4}{3}$$

$$\frac{Y(z)}{z} = \sqrt{3}\left[\frac{2}{3}\left(\frac{1}{z - \frac{1}{2}}\right) + \frac{\left(-\frac{2}{3}\right)z + \left(\frac{4}{3}\right)}{(z^2 - z + 1)}\right]$$

$$Y(z) = \sqrt{3}\left[\frac{2}{3}\left(\frac{z}{z - \frac{1}{2}}\right) + \frac{\left(-\frac{2}{3}\right)z^2 + \left(\frac{4}{3}\right)z}{z^2 - z + 1}\right]$$

$$r = \sqrt{\cancel{p}}$$

$$= 1$$

$$\Omega_0 = \cos^{-1}\left(\frac{-(-1)}{2(1)}\right) = \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{3}$$

Aplicando Z^{-1}

$$y[n] = \sqrt{3}\left[\frac{2}{3}\left(\frac{1}{2}\right)^n u[n] + 1^n\left[\left(\frac{-2}{3}\right)\cos\left(\frac{\pi}{3}n\right) + \frac{2}{\sqrt{3}}\operatorname{sen}\left(\frac{\pi}{3}n\right)\right]u[n]\right]$$

$$A_1 = B = -\frac{2}{3}$$

$$A_2 = \frac{2\left(\frac{4}{3}\right) - \left(-\frac{2}{3}\right)(-1)}{2(1)\operatorname{sen}\left(\frac{\pi}{3}\right)}$$

$$= \frac{\frac{8}{3} - \frac{2}{3}}{\cancel{2}\frac{\sqrt{3}}{\cancel{2}}}$$

$$= \frac{\frac{6}{3}}{\frac{\sqrt{3}}{1}} = \frac{6}{3\sqrt{3}} = \frac{2}{\sqrt{3}}$$

Ánle ...

14. Dado un sistema LTI FIR con respuesta al impulso $h[n] = \{3,2,7,4\}$, obtener su función de transferencia y determinar si el sistema es estable o no.

Recordando la Definición de Transf. $\mathcal{Z}$

$h[n] = 3\delta[n] + 2\delta[n-1] + 7\delta[n-2] + 4\delta[n-3]$

$$\mathcal{Z}\{x[n]\} = X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

por ende:

$$\mathcal{Z}\{h[n]\} = H(z)$$

por ser FIR: $\sum_{n=0}^{3} h[n] z^{-n}$

n: 0 1 2 3

$$h[0]\cancel{z^{0}}^{1} + h[1]z^{-1} + h[2]z^{-2} + h[3]z^{-3}$$

sustituyendo

$$H(z) = 3 + 2z^{-1} + 7z^{-2} + 4z^{-3}\left(\frac{z^3}{z^3}\right)$$

$$H(z) = \frac{3z^3 + 2z^2 + 7z + 4}{z^3}$$

De acuerdo a
La estabilidad en términos de la respuesta al impulso, un sistema es estable:

Continuos

$h(t)$ debe ser absolutamente integrable

Discretas

$h[t]$ debe ser absolutamente sumable

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

como $h[n]$ es finito $L_h = 4$
es absolutamente sumable y

$\therefore$ es ESTABLE